VAHAAN Vehicle Registration Dashboard

Summary

For this assignment, I built a two-dashboard Tableau workbook on the synthesized VAHAN vehicle-registration dataset covering 12 states and roughly 3,700 records from 2018 to 2024. Before building any chart, I engineered eight calculated metrics beyond the raw record counts: Total Registrations (COUNT), Distinct States (COUNTD), Average Vehicle Age (AVG), EV+Hybrid Adoption Rate, BS6 Compliance Rate, 2-Wheeler Share of registrations, a 2018-2024 CAGR, and a 2023-24 Year-on-Year growth rate. These derived metrics let the dashboard speak to volume, composition, pace of change, and the emissions/EV transition at a glance, rather than reporting only raw totals.

The main VAHAAN Overview dashboard opens with a strip of six KPI cards (Total Registrations, EV+Hybrid Adoption, YoY Growth, BS6 Compliance, 2W Share, Average Vehicle Age), followed by the four required charts — Registrations by Vehicle Category, by Fuel Type, by Emission Norm, and by Year — and closes with a record-level Vehicle Detail table. A state dropdown lets the viewer narrow every view to any one of the 12 states, and a companion Key Metrics dashboard serves as a standalone KPI scorecard.

Reflection on Design Decisions

Colour was the decision I spent the most time on. For Fuel Type, I mapped EV and Hybrid to green and left CNG, Diesel, and Petrol in neutral grey, so the eye is pulled straight to the cleaner-fuel share through pre-attentive colour pop-out, instead of five categories competing for attention in a rainbow palette. For Emission Norm, I used a single-hue sequential blue scale, light for BS3 up to dark for BS6, because the norms are ordinal rather than purely categorical, an ordering-preserving choice I deliberately skipped for Vehicle Category, which stayed a neutral, descending-sorted bar chart since those categories carry no inherent rank.

I also ordered the dashboard top to bottom by information density: Key Metrics first, the four categorical and trend charts second, and row-level detail last, following an overview-first, details-on-demand approach rather than burying headline numbers below the charts. Sorting the Vehicle Category bars descending and keeping consistent spacing, alignment, and label placement across all four charts was my way of applying the Gestalt principles of similarity and proximity, so the four panels read as one coherent story rather than four unrelated charts.

For interactivity, I wired all four required charts into one set of filter actions: clicking any bar, segment, or point on the year line cross-filters the other three charts and the Vehicle Detail table below. This gave the drill-down and selection behaviour the brief asked for without needing a separate parameter control, and it lets a viewer go from “what happened nationally” to “which specific vehicles” in two clicks. I also generated an auto phone layout so the same interactions hold on a narrower screen.

If I revisited this dashboard, I would add a reference line or a target band to the CAGR and YoY KPI cards so those numbers had context beyond their own value, and I would test the grey-and-green fuel-type palette against a colour-blindness simulator, since red-green-sensitive viewers may not perceive the intended highlight as clearly as I do.